
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): Coulter Ralph

Student ID: 875 1065

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	7
2	10	10
3	10	10
4	10	10
5	10	10
Total:	50	47

Do not write in the table to the right.

First name: Coulter

Last name: Robt

Student ID: 8751065

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

2
$$\iiint_T f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw$$

□

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

2
$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

□

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

0
$$1 - 2x - 2y = 0 \quad ? \quad \vec{N} = \vec{\nabla} g = \begin{bmatrix} -2x \\ -2y \\ 1 \end{bmatrix}$$

□

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

1
$$\vec{F} = r^2 \cos \theta - 2 \sin \theta \quad ? \quad \times$$

□

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

2
$$\nabla \cdot \vec{F} = \rho^2$$

□

First name: Coulter
 Last name: Zoph
 Student ID: 615 1065

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical

$$h_1 = 1 \quad h_2 = r \quad h_3 = 1$$

$$F = r^2 + z^2$$

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial z} (r(2z)) \right]$$

$$= \frac{1}{r} [4r + 2r]$$

$$= \frac{4r + 2r}{r}$$

$$\nabla^2 f = 6$$

Spherical

$$h_1 = 1 \quad h_2 = \rho \sin \theta \quad h_3 = \rho$$

$$F = \rho^2$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \theta} \left[\frac{\partial}{\partial \rho} (\rho^2 \sin \theta (2\rho)) \right] + \dots$$

$$= \frac{1}{\rho^2 \sin \theta} \left[\frac{\partial}{\partial \rho} (2\rho^3 \sin \theta) \right]$$

$$= \frac{1}{\rho^2 \sin \theta} [6\rho^2 \sin \theta]$$

$$\nabla^2 f = 6$$

□

First name: Louker

Last name: Ralph

Student ID: 875 1065

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\oint_S \vec{F} \cdot \hat{n} dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

$$\iiint_T (\nabla \cdot \vec{F}) dV$$

$$0 \leq \rho \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \pi$$

$$= \int_0^{2\pi} \int_0^\pi \int_0^2 3\rho^2 \sin\theta d\rho d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^\pi -3\rho^2 \cos\theta \Big|_0^\pi d\theta d\phi$$

$$= 3 \int_0^{2\pi} \int_0^\pi [\rho^2] - [-\rho^2] d\theta d\phi$$

$$\nabla \cdot \vec{F} = [x, y, z] \cdot [\partial_x, \partial_y, \partial_z] = 3 \int_0^{2\pi} \int_0^\pi [2\rho^2] d\rho d\theta$$

$$= 3$$

$$= 3 \int_0^{2\pi} \left[\frac{2}{3} \rho^3 \right]_0^2 d\theta$$

$$dx dy dz = \rho^2 \sin\theta d\rho d\theta d\phi = \int_0^{2\pi} 2(\theta) - 0$$

$$= [16]_0^{2\pi}$$

$$\iiint_T (\nabla \cdot \vec{F}) dV = 32\pi$$

□

First name: Coulter

Last name: Zupke

Student ID: 8751065

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Continue problem 4

Solution.

$$\begin{aligned}
 \iint_S \vec{F} \cdot \hat{n} dA \quad n &= [1, 0, 0] \text{ as } r=2 \text{ } \rho=2 \\
 \vec{F} &= [x, y, z] \\
 \vec{F} \cdot \hat{n} &= [\rho] \\
 \iint_S \vec{F} \cdot \hat{n} dA &= \int_0^{2\pi} \int_0^\pi \rho^3 \sin \theta d\theta d\phi \quad \begin{matrix} 0 \leq \theta \leq \pi \\ 0 \leq \phi \leq 2\pi \end{matrix} \\
 &= 8 \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi \quad \rho=2 \\
 &= 8 \int_0^{2\pi} [-\cos(\theta)]_0^\pi d\phi = 8 \int_0^{2\pi} [1] + [1] d\phi \\
 &= 8 \int_0^{2\pi} 2 d\phi \\
 &= 16 [\phi]_0^{2\pi} \\
 &= 16 [2\pi] \\
 \iint_S \vec{F} \cdot \hat{n} dA &= 32\pi
 \end{aligned}$$

□

First name: Louter
 Last name: Ralph
 Student ID: 875 1065

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

$$h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

Solution.

$$\begin{aligned} h_u &= \left| \frac{\partial}{\partial u} \right| = \sqrt{\left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2} \\ &= \sqrt{(ve^u)^2 + (-ve^u)^2} \\ &= \sqrt{v^2 e^{u^2} + v^2 e^{u^2}} \\ h_u &= \sqrt{2v^2 e^{u^2}} \\ h_u &= 2ve^u \end{aligned}$$

$$\begin{aligned} h_v &= \left| \frac{\partial}{\partial v} \right| = \sqrt{\left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2} \\ &= \sqrt{(e^u)^2 + (e^{-u})^2} \\ &= e^u + e^{-u} \\ &= e^u + \frac{1}{e^u} \\ &= \frac{2e^u + 1}{e^u} \end{aligned}$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \right| = \left| \det \begin{bmatrix} ve^u & e^u \\ -ve^u & e^{-u} \end{bmatrix} \right|$$

$$\frac{\partial(x, y)}{\partial(u, v)} = (ve^u)(\frac{1}{e^u}) + (e^u)(v\frac{1}{e^u})$$

$$= v + v$$

$$= 2v$$

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□

First name: Carter

Last name: Zolt

Student ID: 8731063

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$$= \int_0^3 \int_0^u \boxed{22v} \, dv \, du$$

$$= \int_0^3 \left[22 \frac{v^2}{2} \right]_0^u \, du$$

$$= \int_0^3 11u^2 \, du$$

$$= 11 \frac{u^3}{3} \Big|_0^3$$

$$= 11 \cdot 9$$

Triangular Region $2 = \frac{1}{2}$

$$\therefore \text{area} = 4.5 \text{ as } A = 9\left(\frac{1}{2}\right)$$

First name: *Coutter*

Last name: *Rolph*

Student ID: *675 1065*

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Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

$$\iint_A \vec{F} \cdot \hat{n} dA = \iiint_V \nabla \cdot \vec{F} dV$$